

13 HW Never Due, For Final Review Only

Problem 13.1. Determine the convergence or divergence of the following series.

1. $\sum_{n=1}^{\infty} \frac{1}{(2n-1)2^{n-1}};$
2. $\sum_{n=1}^{\infty} \sin \frac{\pi}{2^n};$
3. $\sum_{n=2}^{\infty} \frac{1}{\ln n};$
4. $\sum_{n=1}^{\infty} \frac{\ln n}{n^{\frac{3}{2}}};$
5. $\sum_{n=1}^{\infty} \left(\frac{1+n^2}{1+n^3} \right)^2;$
6. $\sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{1}{\sqrt{n}};$
7. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \ln \left(\frac{n+1}{n-1} \right);$
8. $\sum_{n=1}^{\infty} (e^{\frac{1}{n}} - 1).$ (Hint: Comparison test.)

Problem 13.2. Determine the convergence or divergence of the following series.

1. $\sum_{n=1}^{\infty} \frac{2^n}{(2n-1)!};$
2. $\sum_{n=1}^{\infty} \frac{n^2}{2^n};$
3. $\sum_{n=2}^{\infty} \frac{3^n n}{n^n};$
4. $\sum_{n=1}^{\infty} \frac{(2n-1)!!}{3^n n!};$ (The “double factorial” means we skip by two. So $(2n-1)!! = (2n-1) \times (2n-3) \times \dots \times 1.$)
5. $\sum_{n=1}^{\infty} n^3 \sin \frac{\pi}{3^n};$
6. $\sum_{n=2}^{\infty} 2^{n-1} \tan \frac{\pi}{2^n}.$

Problem 13.3. Determine the convergence or divergence of the following series.

1. $\sum_{n=1}^{\infty} \frac{2^n}{\sqrt{n^n}};$
2. $\sum_{n=1}^{\infty} \frac{1}{3^n} \left(1 + \frac{1}{n} \right)^{n^2};$
3. $\sum_{n=3}^{\infty} \frac{1}{n^p (\ln n)^q (\ln \ln n)^r};$ (The answer depends on the values of $p, q, r.$)
4. $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n+1}} \ln \frac{n+2}{n};$
5. $\sum_{n=1}^{\infty} \left(\sin \left(\frac{\pi}{4} + \frac{1}{n} \right) \right)^n;$
6. $\sum_{n=1}^{\infty} \frac{\ln(n!)}{n!};$
7. $\sum_{n=1}^{\infty} e^{-\frac{n^2+1}{n+1}};$
8. $\sum_{n=1}^{\infty} \frac{(\ln n)^n n!}{n^n};$ (Hint: One method is to use the comparison test, where the key inequality is $n! > \frac{n^n}{e^n}.$
To prove this inequality, note that $e^n = \text{stuff} + \frac{n^n}{n!} + \text{stuff} > \frac{n^n}{n!}.$)
9. $\sum_{n=1}^{\infty} \frac{3n-1}{2^n + 2^{-n}};$
10. $\sum_{n=1}^{\infty} \frac{1}{1+a^n} \quad (a > 0).$ (The answer depends on the value of $a.$)

Problem 13.4. Use Raabe's test to discuss the convergence or divergence of the following series.

1. $\sum_{n=1}^{\infty} \frac{\sqrt{n!}}{(1+\sqrt{1})(1+\sqrt{2})\cdots(1+\sqrt{n})}$;
2. $\sum_{n=1}^{\infty} \frac{n!n^{-p}}{q(q+1)(q+2)\cdots(q+n)}$, $p, q > 0$. (The answer depends on the values of p, q .)

Problem 13.5. Determine whether the following series are absolutely convergent, conditionally convergent, or divergent.

1. $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$;
2. $\sum_{n=1}^{\infty} (-1)^n \frac{(2n-1)!!}{(2n)!!}$; (Hint: Set $A_n = \frac{(2n-1)!!}{(2n)!!}$ and $B_n = \frac{(2n)!!}{(2n+1)!!}$. Then by termwise comparison you can have $A_n < B_n$. Then $A_n^2 < A_n B_n = \frac{1}{2n+1} \rightarrow 0$.)
3. $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1}$;
4. $\sum_{n=2}^{\infty} (-1)^n \sqrt{\frac{n+1}{n(n-1)}}$;
5. $\sum_{n=1}^{\infty} \frac{1}{2^n} \sin \frac{n\pi}{4}$;
6. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^3} \cos \frac{n\pi}{4}$;
7. $\sum_{n=1}^{\infty} (-1)^n \frac{2^{n^2}}{n!}$; (Hint: Standard trick to handle $n!$ is $\frac{n^n}{e^n} < n! < n^n$.)
8. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n - \ln n}$;
9. $\sum_{n=1}^{\infty} (-1)^n (\sqrt{n+1} - \sqrt{n})$;
10. $\sum_{n=1}^{\infty} \sin(\pi\sqrt{n^2+1})$; (Note that $\pi\sqrt{n^2+1} = n\pi + \pi(\sqrt{n^2+1} - n)$. This is actually the Leibniz test/alternating series test!)
11. $\sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n} + (-1)^n}$;
12. $\sum_{n=2}^{\infty} \frac{\cos n\pi}{\sqrt{n} + (-1)^n} \frac{n+1}{n}$;
13. $\frac{1}{\sqrt{2}-1} - \frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}-1} - \frac{1}{\sqrt{3}+1} + \cdots + \frac{1}{\sqrt{n}-1} - \frac{1}{\sqrt{n}+1} + \cdots$; (Also think about why the Leibniz test failed.)
14. $1 - \ln 2 + \frac{1}{2} - \ln \frac{3}{2} + \cdots + \frac{1}{n} - \ln \frac{n+1}{n} + \cdots$. (Set $a_n = \frac{1}{n} - \ln(1 + \frac{1}{n})$ and use Taylor for $\ln(1+x)$.)

Problem 13.6. Find the radius of convergence and the interval of convergence for the following power series.

1. $\sum_{n=1}^{\infty} \frac{x^n}{n^n}$;
2. $\sum_{n=1}^{\infty} \frac{1}{2^n} x^{2n-1}$;
3. $\sum_{n=1}^{\infty} \frac{x^{3n+1}}{(2n-1)2^n}$;
4. $\sum_{n=1}^{\infty} n 4^{n-1} x^{2n}$;
5. $\sum_{n=1}^{\infty} \frac{\ln n}{n} x^n$;
6. $\sum_{n=1}^{\infty} \left[\left(\frac{x}{2}\right)^n + (4x)^n \right]$;
7. $\sum_{n=1}^{\infty} \frac{1}{n^p} (x-1)^n$ ($p > 0$);

8. $\sum_{n=1}^{\infty} \frac{2^{2n-1}}{n\sqrt{n}} (x+1)^n;$
9. $\sum_{n=1}^{\infty} 2^n (x+a)^{2n};$
10. $\sum_{n=1}^{\infty} \frac{(x-a)^{3n}}{(3n)!}.$

Problem 13.7. Find the interval of convergence and the sum function for the following power series.

1. $\sum_{n=2}^{\infty} \frac{x^n}{n(n-1)};$
2. $\sum_{n=1}^{\infty} \frac{1}{4n-1} x^{4n-1};$
3. $\sum_{n=1}^{\infty} (2n+1)x^{2n+1};$
4. $\sum_{n=1}^{\infty} \frac{2n-1}{2^n} x^{2n-2};$
5. $\sum_{n=1}^{\infty} \frac{n(n+1)}{2} x^{n-1};$
6. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n(n+1)} x^n.$

Problem 13.8. Expand the following functions into formal power series at the point x_0 , and find the interval of convergence.

1. $\cos x, \quad x_0 = \frac{\pi}{4};$
2. $\sin\left(x + \frac{\pi}{6}\right), \quad x_0 = 0;$
3. $\ln(1+x), \quad x_0 = 2;$
4. $\cosh x, \quad x_0 = -1;$
5. $\sin x^2, \quad x_0 = 0;$
6. $\ln(3-x), \quad x_0 = -1;$
7. $\frac{1}{x-1}, \quad x_0 = -1;$
8. $\frac{x+2}{x^2-x-2}, \quad x_0 = -2;$
9. $\frac{x}{(x-1)(x+3)}, \quad x_0 = 0;$
10. $\frac{x}{2x-1}, \quad x_0 = -1;$
11. $\ln(x + \sqrt{x^2+1}), \quad x_0 = 0;$
12. $\frac{2}{(1-x)^3}, \quad x_0 = -1;$
13. $\int_0^x \frac{\arctan t}{t} dt, \quad x_0 = 0;$
14. $\int_0^x \frac{1}{\sqrt{1+t^3}} dt, \quad x_0 = 0.$

Problem 13.9. Expand the following functions into formal Fourier series.

1. $f(x) = \begin{cases} x + \pi, & x \in [-\pi, 0), \\ \pi - x, & x \in [0, \pi]; \end{cases}$
2. $f(x) = |\sin x|, \quad x \in [0, 2\pi];$
3. $f(x) = \frac{\pi}{4} - \frac{x}{2}, \quad x \in (-\pi, \pi);$

4. $f(x) = \begin{cases} 1, & x \in [0, h], \\ 0, & x \in (h, \pi) \end{cases}$ expand into a cosine series (even function) with period 2π ;
5. $f(x) = x + x^2, \quad x \in [0, 2\pi]$ expand into a Fourier series with period 2π .

Problem 13.10. For the following differential equations (or integral equation), compute the Fourier coefficients of any solution $f \in \mathcal{C}$, and find all solutions in $\mathcal{C}$.

1. $f'(x) + f(x) = \sin(2x)$.
2. $f(x + \pi) = f'(x + 1)$.
3. $f(x) + \int_0^x f(t) dt = 1 - \cos x + \sin x$. We assume $\hat{f}_0 = 0$. (Hint: Use Fubini to compute the Fourier coefficients of $\int_0^x f(t) dt$ in terms of the Fourier coefficients of $f(x)$.)

Problem 13.11. Find the value of the following series, which is related to the Fourier series of the given function.

1. $\sum_{n=1}^{\infty} \frac{1}{n^2 + a^2}, f(x) = e^{ax}$ on $[-\pi, \pi]$.
2. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + a^2}, f(x) = e^{ax}$ on $[-\pi, \pi]$.
3. $\sum_{n=1}^{\infty} \frac{1}{n^2 - a^2}, f(x) = \cos(ax)$ on $[-\pi, \pi]$ and a is not an integer.
4. $\sum_{n=1}^{\infty} \frac{1}{(4n^2 - 1)^2}, f(x) = |\sin x|$.

Problem 13.12 (Less Important). Determine the convergence or divergence of the following series, and find the sum of the convergent series.

1. $\sum_{n=1}^{\infty} 100 \left(\frac{1}{4}\right)^{n-1}$;
2. $\sum_{n=1}^{\infty} \sqrt{\frac{n+1}{n+2}}$;
3. $\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+3)}$; (Hint: Partial fractions. Same trick for several cases below.)
4. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$;
5. $\sum_{n=1}^{\infty} \frac{(-1)^n n^3}{2n^3 + n}$;
6. $\sum_{n=1}^{\infty} (\sqrt{n+2} - 2\sqrt{n+1} + \sqrt{n})$;
7. $\sum_{n=1}^{\infty} \arctan \frac{1}{2n^2}$; (Hint: $\frac{1}{2n^2} = \frac{(2n+1)-(2n-1)}{1+(2n+1)(2n-1)}$, and compare this with the formula for $\tan(x-y)$.)
8. $\sum_{n=1}^{\infty} \frac{2n-1}{2^n}$; (Hint: Prove absolute convergence first. Then let S be the result, and consider the cancellation in the terms of $S - \frac{1}{2}S$.)
9. $\sum_{n=1}^{\infty} \frac{1}{\sqrt[n]{n}}$.
10. $\sum_{n=1}^{\infty} \frac{1}{n(n+m)} \quad (m > 0, m \in \mathbb{Z})$.

Problem 13.13 (Less Important). Discuss the convergence or divergence of the following series.

1. $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \sqrt{\ln \frac{n+1}{n}} \right)$; (Hint: Use Taylor approximation of the correct degree for $\ln(1+x)$.)
2. $\sum_{n=1}^{\infty} (n^{\frac{1}{n^2+1}} - 1)$. (Hint: $n^{\frac{1}{n^2+1}} = e^{\frac{\ln n}{n^2+1}}$. Then use Taylor approximation of the correct degree for $e^x - 1$.)

Problem 13.14 (Less Important). Discuss for what values of p , the series $\sum_{n=2}^{\infty} \frac{(-1)^n}{[n+(-1)^n]^p}$ and $\sum_{n=2}^{\infty} \frac{(-1)^n}{[\sqrt{n+(-1)^n}]^p}$ respectively converge absolutely, converge conditionally, or diverge.

Problem 13.15 (Less Important). ♣ For what values of p does the series $\sum_{n=1}^{\infty} \frac{\sin n}{n^p + \sin n}$ converge absolutely, converge conditionally, or diverge. (Hint: For $p > 1$ this is easy. For $0 < p \leq 1$, note that $\frac{\sin n}{n^p + \sin n} = \frac{\sin n}{n^p} (1 + \frac{\sin n}{n^p})^{-1}$ and use Taylor for $(1+x)^{-1}$. You will see a term with $(\sin x)^2$ on the numerator, i.e., that portion is NOT alternating!)

Problem 13.16 (Less Important). Prove the following identities.

1. $\prod_{n=2}^{\infty} \frac{n^3-1}{n^3+1} = \frac{2}{3}$; (Hint: Factorize $n^3 - 1$ and $n^3 + 1$. Then write a few terms to see the cancellation.)
2. $\prod_{n=2}^{\infty} \left(1 - \frac{2}{n(n+1)}\right) = \frac{1}{3}$; (Hint: Note that $1 - \frac{2}{n(n+1)} = \frac{n^2+n-2}{n(n+1)}$. Factorize, write a few terms to see the cancellation.)
3. $\prod_{n=0}^{\infty} \left(1 + \left(\frac{1}{2}\right)^{2^n}\right) = 2$. (Hint: Note that

$$(1-x)(1+x)(1+x^2)(1+x^4)\dots(1+x^{2^N}) = 1 - x^{2^{N+1}}.$$

Magic indeed.)

Problem 13.17 (Less Important). Discuss the convergence of the following infinite products.

1. $\prod_{n=2}^{\infty} \frac{1}{n}$;
2. $\prod_{n=1}^{\infty} \frac{(n+1)^2}{n(n+2)}$;
3. $\prod_{n=1}^{\infty} \sqrt[n]{1 + \frac{1}{n}}$;
4. $\prod_{n=1}^{\infty} \frac{n}{\sqrt{n^2+1}}$;
5. $\prod_{n=2}^{\infty} \left(\frac{n^2-1}{n^2+1}\right)^2$;
6. $\prod_{n=2}^{\infty} \sqrt[n]{\ln\left(1 + \frac{x}{n}\right)}, \quad x \neq 0, -1, -2, \dots$